

Magma Series & Plate Tectonic Settings — Complete Concept Guide

Alkaline · Tholeiitic · Calc-Alkaline · Classification Diagrams & Tectonic Distribution

Reference: Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed. — Chemical Petrology I

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1. What Is a Magma Series?

Magma series: A group of igneous rocks that share chemical (and often mineralogical) characteristics and show a consistent evolutionary trend on a variation diagram, implying they descended from a common parental magma via a shared evolutionary process.

"A group of rocks that share some chemical (and perhaps mineralogical) characteristics and shows a consistent pattern on a variation diagram, suggesting a genetic relationship, can be referred to as a magma series. Synonymous terms include association, lineage, magma type, and clan."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 158

The concept traces back to Iddings (1892), who first proposed that igneous rocks fall into either an **alkaline** or a **subalkaline** series, each evolving from a distinct type of parental magma toward more silicic derivatives. Harker (1909) later observed the same pattern geographically, dividing Cenozoic circum-oceanic volcanics into an "Atlantic" (alkaline) branch and a "Pacific" (subalkaline) branch.

2. The First Split — Alkaline vs. Subalkaline

"Alkaline rocks are silica undersaturated, whereas subalkaline rocks are silica saturated to oversaturated."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 158

This split is visualized on a **Total Alkali-Silica (TAS) diagram** — weight % ($\text{Na}_2\text{O} + \text{K}_2\text{O}$) plotted against weight % SiO_2 :

"The alkaline and subalkaline series are distinguished in a total alkali versus silica diagram... the alkaline rocks plot distinctly above and the subalkaline rocks below the dividing lines chosen by MacDonald (1968) and Irvine and Baragar (1971)."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 159

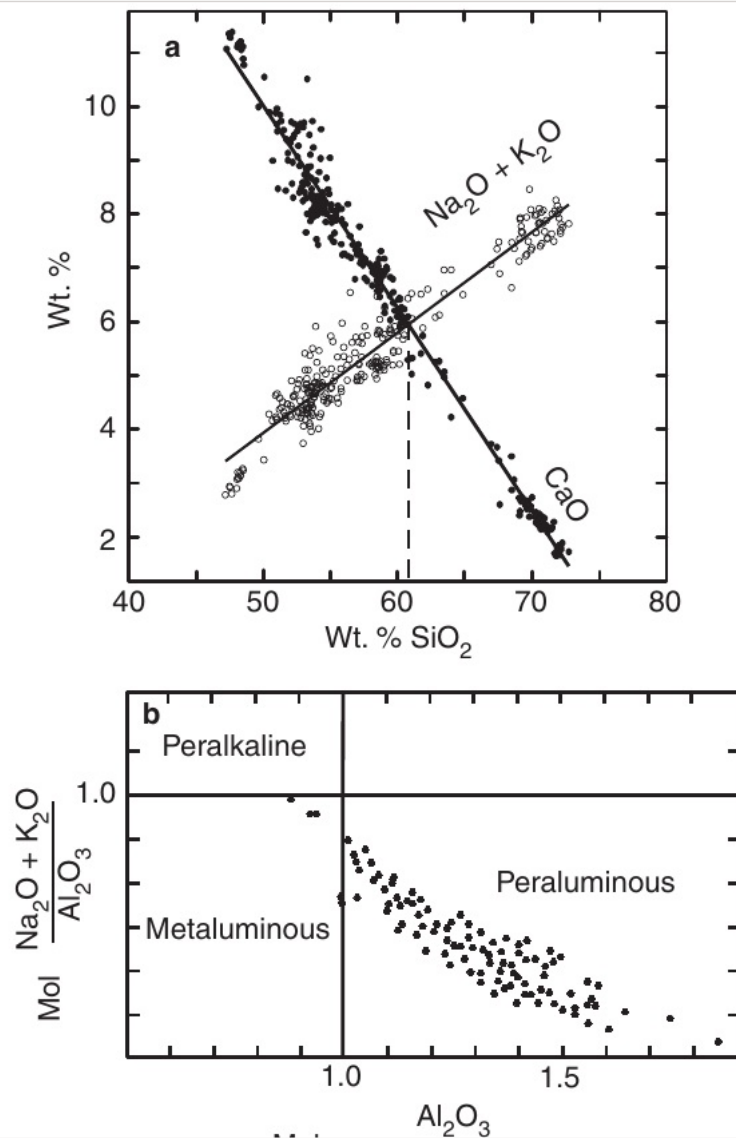


Figure 11 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 159) — Total alkalis versus silica diagram for the alkaline (open circles) and subalkaline rocks of Hawaii. The alkaline field sits clearly above the subalkaline field at any given SiO_2 content.

Evolutionary pathway difference

"The common evolutionary sequence in the alkaline series begins with an alkali olivine basalt and proceeds through trachybasalts and trachyandesites, to trachytes or phonolites." Subalkaline magmas instead evolve toward either quartz-normative (silica-saturated) or olivine-normative compositions across the "thermal divide" at the albite composition, which prevents the two series from crossing into one another as they cool (p. 159-160).

3. The Second Split — Tholeiitic vs. Calc-Alkaline

"Most investigators continue to recognize Iddings's (1892a) original two series, alkaline and subalkaline, and accept that the subalkaline can be further divided into the tholeiitic series and calc-alkaline series."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 159

"The subalkaline series was further subdivided into a tholeiitic and a calc-alkaline series by Tilley (1950)."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 160

Interestingly, these two series **cannot** be told apart on the alkali-silica or Ne-OI-Q diagrams used to separate alkaline from subalkaline — they need different, dedicated diagrams (Section 4).

4. Classification Diagrams: TAS, AFM, and Al_2O_3 -Plagioclase

Because tholeiitic and calc-alkaline trends overlap on a TAS diagram, petrologists instead use an **AFM diagram** (Alkalis-FeO(total)-MgO ternary) to separate them:

"If we compare on AFM diagrams... we see that the Skaergård trend is clearly tholeiitic, and the Crater Lake trend is calc-alkaline. Both series progress along the basalt-andesite-dacite-rhyolite trend, but there are distinctive mineralogical and chemical differences between the two series that are most evident for intermediate compositions."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 160

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both CaO and alkalis (as the original Peacock criteria do).

Irvine and Baragar (1971) further divided the alkali series into a **sodic series** and **potassic series**, and Middlemost (1975) recommended adding a **high-K series** as well (Figure 16). The remaining terms, such as *peraluminous*, *metaluminous*, etc., although they may yet be shown to characterize some distinctive magma series, are more useful as descriptive terms when some chemical characteristic is emphasized for a particular igneous rock or province.

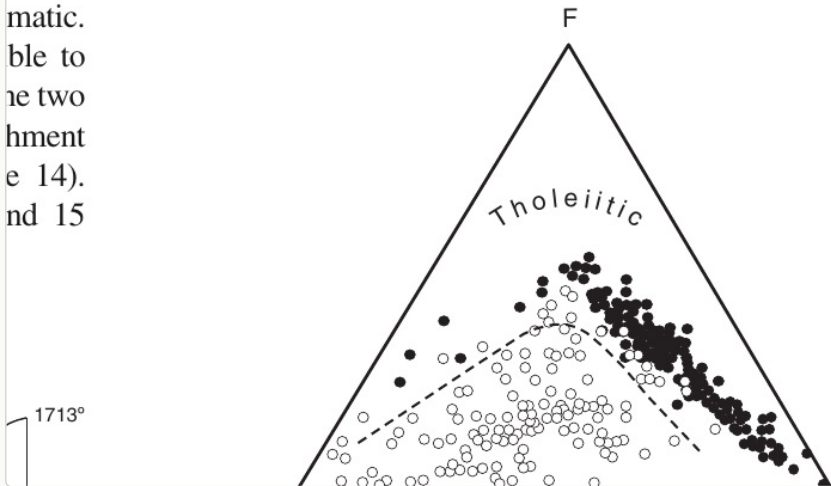


Figure 14 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 160) — AFM diagram distinguishing tholeiitic rocks (Iceland, Mid-Atlantic Ridge, Columbia River Basalts, Hawaii — solid circles) from calc-alkaline rocks of the Cascade volcanics (open circles). Tholeiitic liquids show strong iron enrichment before differentiating toward silica; calc-alkaline liquids differentiate toward silica-rich compositions without significant Fe-enrichment.

A second diagnostic diagram plots wt.% Al₂O₃ against the anorthite content of normative plagioclase (Winter's Figure 15, Irvine & Baragar 1971) — tholeiitic and calc-alkaline rocks again plot as distinct fields, useful when AFM data isn't available.

5. Magma Series and Plate Tectonic Setting

This is the section that directly answers Q.19. Winter cross-tabulates each series against every major tectonic setting:

cons of magmas at mid-ocean ridges. Magma series are important to our understanding of igneous petrogenesis. Earlier, as we first regarded the spectrum of igneous rocks that occur on Earth, I asked a basic	were several parents, each with its own range of differentiation products? It would appear that the latter is the case and that there are at least three principal magma series, and perhaps several other minor ones.
Summary Petrologists use the chemical composition of rocks and minerals to characterize and categorize individual types, to identify series of related rocks, and to model processes that relate the rocks in a consanguineous series. Analyses were initially performed by using tedious gravimetric methods but are now quickly and routinely accomplished by using instrumental spectroscopic techniques based on emission or absorption of energy when directed at prepared samples. Analyses are typically reported in wt. % oxide for the major	used, particularly ones in which various components are plotted versus some chemical parameter that is considered a measure of chemical fractionation (e.g., SiO₂, Zr, Mg#). Ternary diagrams, such as the AFM diagram, are also common. By using bivariate diagrams, such as Harker or Pearce element ratio diagrams, we can evaluate specific differentiation mechanisms (e.g., fractional crystallization of certain minerals) deemed responsible for the chemical evolution within a suite of supposedly consanguineous rocks. Quanti-

Table 7 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 162, after Wilson 1989) — Occurrence of the alkaline, tholeiitic, and calc-alkaline series

Calc-Alkaline — Convergent Margins ONLY

"First, calc-alkaline magmas are essentially restricted to subduction-related plate tectonic processes. This must carry some genetic significance and will certainly be a factor in any model for subduction zone magma genesis. Recognition of calc-alkaline characteristics in the composition of ancient volcanic rocks may help determine their tectonic framework."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 161

Repeated later: "Because the calc-alkaline series is restricted mostly to convergent plate boundaries in complex settings, we shall set it aside [when modeling oceanic basalt generation]." (p. 191)

Tholeiitic — Everywhere, Dominant at Divergent Margins

"Second, tholeiitic magmas are practically the exclusive magma type associated with divergent boundaries. Although some alkaline rocks are found, they play a subordinate role, usually in the initial stages of continental rifting."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 161

Tholeiitic basalts (as MORB) are generated at mid-ocean ridges and are also found at oceanic islands, convergent margins, and continental settings — the most volumetrically dominant and tectonically widespread series of the three.

Alkaline — Widespread Except at Divergent Margins

Per Table 7: found at convergent margins, oceanic within-plate settings (classic ocean-island alkaline basalts), and continental within-plate settings (continental rifting, intraplate volcanism) — but essentially absent as a dominant type at divergent (mid-ocean ridge) margins, where tholeiitic dominates instead.

6. Komatiite — A Time-Restricted, Not Tectonically-Restricted, Rock Type

Komatiite (glossary definition): "Ultramafic volcanic (mostly Archean): wt.% MgO > 18%, TiO₂ < 1%." (Winter, p. 34)

"Komatiites, ultramafic volcanic rocks, are almost completely restricted to the Archean, when the geothermal gradient was much higher, and a larger fraction of partial melting (F) was apparently easier to generate."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 198

Why this matters for Q.19

Komatiite is defined by **composition and geologic age** (an ultramafic lava essentially unique to the hotter Archean mantle), not by plate tectonic setting. It is not one of the three principal magma series (alkaline/tholeiitic/calc-alkaline) at all, so it cannot be "restricted to subduction zones" in the sense the question asks.

7. Q.19 Worked Solution

Q.19. Which one of the following magma series is found only in subduction zones?

- (a) Calc-alkaline
- (b) Tholeiitic
- (c) Alkaline
- (d) Komatiite

Option	Tectonic Distribution (per Table 7)	Verdict
(a) Calc-alkaline	Convergent only — the single "yes" in the whole table	Correct
(b) Tholeiitic	Convergent, divergent, oceanic, and continental — the <i>most</i> widespread series	Eliminated
(c) Alkaline	Convergent, oceanic, and continental (absent only at divergent margins)	Eliminated
(d) Komatiite	Not a tectonic-setting category; an Archean-age, MgO-rich ultramafic lava type	Eliminated

✓ Answer: (a) Calc-alkaline

8. Quick-Reference Summary & Probable Exam Questions

Series	Silica Saturation	Key Diagram(s)	Tectonic Setting
Alkaline	Undersaturated	TAS (Fig. 11)	Convergent, oceanic & continental within-plate (not divergent)
Tholeiitic	Saturated-oversaturated	AFM (Fig. 14), Al ₂ O ₃ -plag (Fig. 15)	All settings; dominant at divergent margins
Calc-alkaline	Saturated-oversaturated	AFM (Fig. 14), Al ₂ O ₃ -plag (Fig. 15)	Convergent (subduction) only

Q. Which two series cannot be distinguished on a TAS or Ne-Ol-Q diagram, and what diagram must be used instead?

A. Tholeiitic and calc-alkaline; distinguished instead via the AFM diagram or an Al₂O₃ vs. normative-plagioclase plot.

Q. What geochemical behavior distinguishes tholeiitic differentiation from calc-alkaline differentiation on an AFM diagram?

A. Tholeiitic liquids show strong early iron enrichment before evolving toward silica-rich compositions; calc-alkaline liquids evolve toward silica-rich compositions without significant iron enrichment.

Q. What is the "thermal divide," and what does it prevent?

A. A compositional barrier at the albite composition on the Ne-Q system that prevents alkaline and subalkaline liquids from crossing into each other's fields as they cool — keeping the two series distinct.

Q. Who first proposed the alkaline/subalkaline distinction, and who first subdivided subalkaline into tholeiitic and calc-alkaline?

A. Iddings (1892) proposed alkaline/subalkaline; Tilley (1950) subdivided subalkaline into tholeiitic and calc-alkaline.

Q. Why can't komatiite be considered a "magma series" in the same sense as alkaline/tholeiitic/calc-alkaline?

A. It's defined by a compositional threshold (MgO > 18%, TiO₂ < 1%) and near-total restriction to the Archean, not by a distinct evolutionary lineage tied to a tectonic setting.

Q. At which tectonic setting is tholeiitic magma overwhelmingly dominant, with alkaline playing only a subordinate role?

A. Divergent boundaries (mid-ocean ridges); alkaline rocks appear there only in minor amounts, usually during early continental rifting.